DID for Secure AIS Messages and Data Delivery for Maritime Vessels

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Abstract. Advanced maritime communication supports the use of satellite infrastructures for low-latency services. In recent years, Low Earth Orbit (LEO) satellite-based services have been explored as a viable option for shore-to-ship communication. However, besides affordable and low-latency services, the operation is sometimes affected because of extreme weather conditions and geographic censorship of LEO constellations.

In such scenarios, proximity-based routing supporting ad-hoc data transmissions can be an alternative. Proximity information is obtained from the vessel's Automated Identification System (AIS) messages. AIS is a standard broadcast service used for vessel identification that relies on the vessel's GNSS receivers. Because of their unauthenticated nature, GNSS and AIS messages are prone to spoofing, man-in-the-middle, or replay attacks. Protection against rogue vessels flooding AIS messages remains a challenge. Furthermore, vessels currently lack an accountability mechanism for their reported coordinates. Consequently, a vessel with malicious intent or one that has been compromised may potentially provide false positional information, thereby disrupting maritime operations.

Solutions such as Galileo OSNMA provide signal authentication but fail to assure the signal origin. To address this issue, the Signal Authentication Service (SAS) is introduced, which assures the origin of the signal is a Galileo satellite. However, for the SAS to work, it requires certain artifacts to be made available in advance before verification can begin. These artifacts are time-bound and their generation beyond a certain timeframe is considered to be insecure. The delivery of these artifacts to vessels operating offshore is a challenge. At the same time, direct dependency on the Galileo GSC server is a privacy risk. Thus, a secure decentralized data delivery system is required.

In this work, we propose an AIS-based data delivery system. We use a decentralized identity (DID) to authenticate AIS messages, provide replay protection, and the delivery of SAS artifacts. This work relies on LEO constellations for data delivery and falls back to peer-to-peer data sharing when needed.

Keywords: AIS · Decentralized Identity · Galileo GNSS · LEO

1 Introduction

The Global Navigation Satellite System (GNSS) has allowed maritime vessels to precisely locate themselves in the ocean. GNSS comprises constellations of

Earth-orbiting satellites that broadcast their location in space. GNSS receivers capture these relayed signals and calculate ground positions. GNSS not only allows the vessel to calculate its position, but at the same time it allows the vessel to determine its navigation, velocity, and time. Maritime self-reporting systems such as the Automated Identification System (AIS) heavily depend on GNSS for calculating Position, Navigation, and Timing (PNT) information in order to transfer it. Despite its usefulness, ensuring correct calculation of the parameters from the received signals from the satellites is a challenge. The GNSS signals transmitted from space can be received by any compatible GNSS receivers; this exposes the scope for different sorts of attacks. An attack can capture a GNSS signal and can later relay the captured signal, making the legitimate receiver on the vessel calculate valid but incorrect parameters. There is a lack of an authentication mechanism in most of the GNSS systems. Addressing these concerns is of prime importance, as such attacks can result in catastrophes such as accidents. Like the position values obtained from the received signals are shared with nearby vessels over AIS messages for hull-to-hull docking. Few works have been done to address such sorts of challenges; for example, Galileo, yet another GNSS system, attempts to solve this problem by providing Open Service Navigation Message Authentication (OSNMA), an authentication mechanism to its relayed signals. OSNMA enables the receiver to verify that a signal is not outdated or modified beyond a certain timeframe. However, it is still unable to address spoofing and meaconing attacks. In recent years, Galileo released the Signal Authentication Service (SAS). SAS ensures that the signal received by a receiver is coming directly from one of the Galileo satellites, not from any other entity. This completely rules out the scope of signal spoofing. However, there is a prerequisite for the SAS to work. SAS requires the receiver to possess certain artifacts called reencrypted code sequences (RECS) before the signal verification can take place. These RECS files are made available through the Galileo servers; receivers can download these artifacts in advance but no longer than 7 days in the future. This condition can seem acceptable for on-ground receivers having access to the internet but is a bottleneck for offshore vessels sailing in the middle of the oceans. Long-haul vessels, such as transportation vessels, can be in the ocean for weeks, away from the internet infrastructure. Storing RECS files for more than 7 days of verification is considered to be insecure[5]. Moreover, periodic download of RECS files from the Galileo servers can result in behaviour profiling of the vessels and can be a concern for the shipping businesses. There are reported incidents of service disruptions because of Galileo server unplanned downtime[11]. This calls for a decentralized, secure, and privacy-preserving approach for RECS file dissemination.

In this line, this paper introduces a satellite-assisted cum ad-hoc peer-to-peer data transfer mechanism for maritime vessels.

2 Related Works

AIS is now a fundamental part of maritime communications and is used for applications including, but not limited to, collision avoidance, self-reporting, maritime surveillance, and voyage safety [1]. [4] proposes an AIS-based routing protocol to extend shore-based networks to near-shore waters. An opportunistic network routing protocol for maritime voluntary observation vessels is provided in [14], offering another routing approach for maritime networks. AIS supports a number of message types, each carrying different information. However, because these messages are unencrypted and unauthenticated, they are prone to attacks such as replay and spoofing [3].

Reference	**AuthAIS**[2]	**SecAIS** [12]	**Su et.al** [18]	**Our Work**
Authenticated	✓	✓	✓	✓
Replay Protection	×	×	×	✓
Decentralised	×	×	×	✓
Self-Sovereign	×	×	×	✓
Privacy Preserving	✓	✓	✓	✓
Location Correctness Gurantees	×	×	×	✓

Table 1. Comparative Analysis of the Work

Several works study security challenges in AIS [17]. To counter these challenges, several research efforts have been made over the years. In [18], the authors proposed a privacy-preserving vessel authentication scheme using an automatic identification system. It relies on a digital certificate-based identity authentication scheme for providing authenticity to AIS messages and a mixzone and blind signature-based trajectory privacy-preserving scheme. One issue associated with this approach is that it relies on a centralized trust authority, i.e., the Certificate Authority (CA). Unavailability of the CA may make the entire approach ineffective. A secure automatic identification system (SecAIS) is discussed in [12]. There also exist a couple of works that use identity-based cryptography for secure AIS [13]. Auth-AIS: Secure, Flexible, and Backward-Compatible Authentication of Vessels AIS Broadcasts [2] utilises TESLA and Bloom Filters for authentication. Security is provided using an HMAC-based approach; however, because it uses HMAC with a symmetric key, it raises several issues. If the same key is used between two vessels, it would be difficult to determine which one is the sender and which one is the receiver. If different keys are used for the sender and the receiver, then there will be issues associated with key management. Moreover, it needs a secure mechanism to deliver the key to each vessel before the communication takes place, which is not a scalable idea. Table 1 provides a comparative analysis of the challenges unaddressed by the existing works.

Moreover, we have not found any work in the literature that attempts to bring confidence in the PNT calculation made by the vessel before it gets transmitted as part of the AIS message.

3 Motivation

Despite several efforts in recent years, there is a lack of scalable, secure, and privacy-preserving mechanisms for AIS message delivery. Existing works mainly rely on centralised infrastructure, which is neither reliable nor highly available. Decentralised technologies such as the blockchain and decentralised identity can be a possible contender for a secure and privacy-preserving AIS mechanism. There has been sufficient work done where DID has been used for seamless control, especially in IoT. Building upon these decentralised technologies, we in this work aim to address the challenges mentioned in [15]. In short, the goals of this work are as follows:

1. Facilitate location correctness with an artifact delivery mechanism for Galelio SAS for geographic location validity checks
2. Provide an order check for AIS messages
3. Provide a secure and privacy-preserving authentication mechanism for AIS messages
4. Provide integrity checks for AIS messages

In addition to the above-mentioned goals, the work provides the vessel with self-sovereign control over its data. This can be extremely useful while interacting with non-trusting entities such as border security in international waters. The vessel can decide which information to disclose based on the requirement and then disclose that specific information only.

4 Background

This section provides the necessary background for our work.

4.1 Galileo OSNMA

On July 24, 2025[7], the Galileo Open Service Navigation Message Authentication (OSNMA) service became active, becoming the first civilian GNSS constellation to offer cryptographic authentication of its navigation signals. OSNMA employs the TESLA (Timed Efficient Stream Loss-tolerant Authentication) protocol[8], making signal forgery computationally infeasible for an attacker, as any receiver that implements the full TESLA key chain verification signal authenticity can easily verify signal authenticity. This represents a decisive advance against the dominant class of GNSS spoofing attacks.

OSNMA authenticates Galileo I/NAV navigation messages using the TESLA protocol [8]. A one-way key chain K[0],K[1],. . . ,K[N] is generated such that K[i]=H(K[i+1]), where H is SHA-256 truncated to 128 bits. Each key K[i] is used to generate HMAC authentication tags for navigation messages transmitted during epoch i and is disclosed two epochs later (a 60-second delay, as specified in ICD v1.1 [8]). Receivers verify the integrity of the disclosed keys by checking the one-way chain relation K[i]=H(K[i+1]), anchored by the ECDSA-P256-signed

root key (KROOT) distributed by EUSPA. Galileo I/NAV messages are transmitted in 30-second subframes, each comprising fifteen 2-second pages. Because the TESLA key schedule is defined globally, honest receivers observing the same Galileo System Time (GST) epoch are expected, under nominal reception conditions, to validate the same disclosed TESLA key and accumulate authentication tags generated with the same current key for commonly observed satellites.

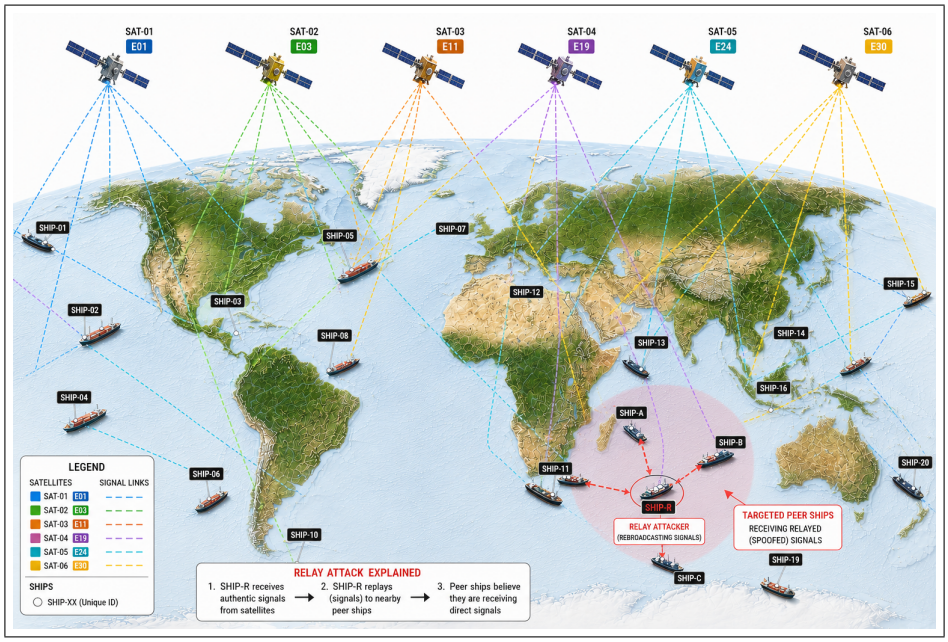


Fig. 1. Global view of OSNMA signal reception and meaconing attack: *6 Galileo satellites (E01, E03, E11, E19, E24, E30) broadcast authenticated signals received by multiple ships. In the ocean, SHIP-R acts as a meaconing attacker; it receives authenticated signals from the satellite and rebroadcasts them to nearby ships, which may accept the relayed signals.*

However, it is acknowledged — in the OSNMA ICD itself and in the academic literature — to not protect against meaconing (relay) attacks[20][16]. A relay attack involves retransmitting genuine signals that are authenticated by OSNMA from an actual location to a receiver located elsewhere. Since the signals are legitimate, the OSNMA verification is successful. Fig. 1 shows a scenario of a relay attack in the ocean. The crisis in the Strait of Hormuz in 2026, where more than 1,100 vessels faced synchronized GPS disruptions within a single day [9], along with the 1,900 recorded GPS interference incidents at eight airports in India from 2023 to 2025 [6], highlights the pressing need to address this vulnerability. Current methods for detecting relay interference are either based on single-receiver strategies (such as clock offset anomaly, Doppler consistency, and

IMU divergence [19]) that adept attackers can circumvent or multi-receiver geometric techniques (like TDOA double differences) that necessitate the physical closeness of receivers. Neither approach is suitable for a globally distributed fleet, such as a naval organisation or a shipping company, where receivers are situated across vast ocean regions.

4.2 Galileo SAS

Fig. 1 shows how a replay attack can be performed even if an authenticated GNSS service like OSNMA is used. In order to address the challenges associated with OSNMA, a signal authentication service (SAS) is proposed to be introduced. SAS complements OSNMA by proving the signal received by the GNSS receiver originates from the satellite of Galileo constellation. Galileo SAS uses the existing Galileo E6C signal to be encrypted, in combination with OSNMA. Portions of E6C are re-encrypted with the OSNMA future keys and are published on a server[10]. These Re-Encrypted Code-Sequences (RECS) are published on the GSC servers, from where the receivers can download these.

4.3 Automated Identification Service (AIS)

AIS plays a crucial role in traffic management, search and rescue, and collision avoidance[15]. AIS is a situational awareness system that allows vessels at sea and maritime authorities to be aware of their presence by broadcasting identification and position information. AIS was conceived in the early 1990s particularly for collision avoidance on the availability-only feature, and security was never considered. AIS messages are broadcast in plain text without any encryption, signature, and hashes, allowing any malicious entity to capture the broadcast message with a Software Defined Radio (SDR). The captured messages can later be forged and replayed by the malicious entity, deceiving the vessel into changing its course. A malicious entity, such as a group of pirates, can perform capture and replay of AIS messages to force a shipping vessel to enter a vulnerable area. Each AIS device is identified by a unique Maritime Mobile Service Identity (MMSI). Although the MMSI is considered to be unique and specific to a vessel, because of the absence of any authentication support in AIS, there are no guarantees. These MMSI numbers are prone to forgery and tampering. There are reports where the same MMSI is shared across several vessels at different time instances[18].

5 Proposed System

This section describes the system model, the attack model, the authentication layer, and DID management in detail.

5.1 System Model

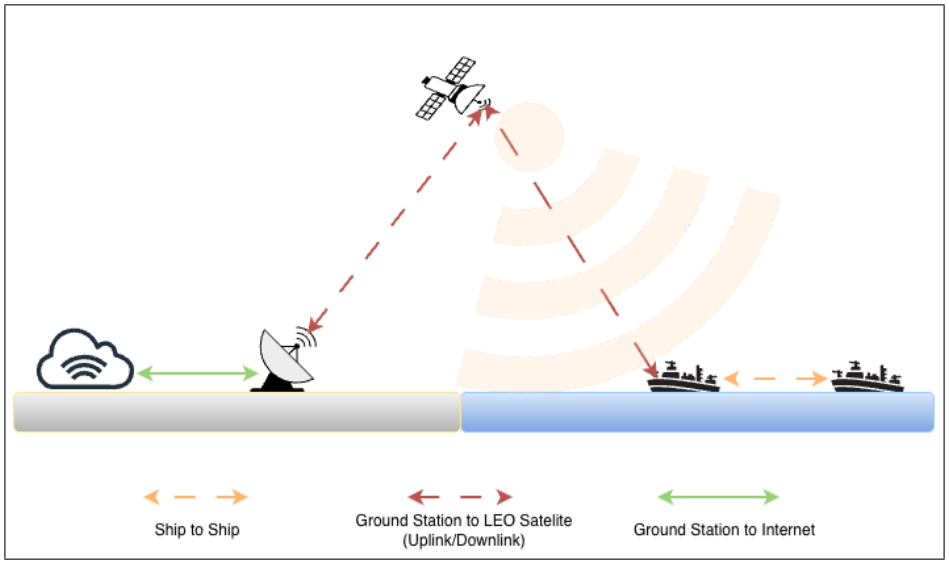


Fig. 2. System Model

Our system model, as shown in the Figure 2, consists of three different entities: the vessels, the LEO satellites, and the shore-based ground stations. Each vessel has an OSNMA receiver, a SAS processing module, and AIS installed. Each vessel is uniquely identified with a DID. The private key associated with the DID is securely stored on the vessel and remains under the sole control of the vessel. Vessels can now digitally sign the AIS messages and encrypt communications with their peers. To sign the AIS messages, the vessel uses its private key; the peer vessels, upon receiving the signed message, verify it against the public key of the signing vessel. The DID registration, revocation, and resolution process is described in section 5.4.

5.2 Vessel Capabilities

1. *OSNMA Receiver Module*: It receives and processes Galileo E1-B signals, verifies TESLA authentication chains, and maintains local authentication-state evolution.
2. *SAS Processing Module:* Software capable of processing encrypted E6-C signals for generating anti-spoofing and anti-replay PVT data.
3. *Secure TPM:* A trusted platform module responsible for storing private keys securely. It stores the private key of the DID in our case.

4. *Communication Interface:* exchanges telemetry with fleet infrastructure. Data can be transferred over maritime networks—including VDES, AIS extensions, Wi-Fi mesh, LTE/5G, satellite links, and delay-tolerant opportunistic networks—while remaining resilient to packet loss, communication delays, network partitions, and the absence of shore-side infrastructure.

5.3 LEO Satellite

LEO satellites provide low-latency data delivery as they fly at relatively low altitudes (typically between 200 and 2000 kms.)[21]. In our system, a LEO satellite is provided with a DID registered on the blockchain. The key pair associated with the DID is used for authentication. The LEO satellite talks to the maritime server hosting the artefacts through the ground station. The maritime server in turn fetches the latest artefacts periodically from the GSC via the internet. The LEO satellite is also responsible for delivering updated Merkle Root information to the offshore vessels.

5.4 Decentralized Identity

DID Management While Galileo Open Service Navigation Message Authentication (OSNMA) guarantees the authenticity and integrity of navigation messages transmitted by Galileo satellites, it does not authenticate the identity of vessels exchanging observations. To establish trust among collaborating vessels, we need a robust, decentralized authentication mechanism with high availability. Decentralized Identity (DID) is a strong candidate.

A DID is a globally unique cryptographic identifier anchored preferably on a distributed ledger such as a blockchain. For a vessel, it is issued by the flag state authority at registration—analogous to the MMSI, but unlike the MMSI, it is cryptographically bound to a key pair stored on the vessel. The vessel's DID document, as shown in Fig. 4, is publicly resolvable by anyone on the ledger and contains the vessel's public key, communication endpoints, and other non-PII, etc. It is a machine-verifiable identity card for the vessel.

A DID-based approach provides several security benefits:

1. Authentication: Confirms the identity of the vessel producing each observation.
2. Authorization: Ensures that only credentialed vessels participate in the system.
3. Non-Repudiation: Cryptographically binds every transmitted observation to its originating vessel.

The vessel DID is bound to the vessel's hardware, which makes it immune to identity theft, as the only compromise of the keys will not allow the attacker to sign messages on behalf of the vessel. A successful attack would instead require compromise of the vessel's hardware, for which physical access to the vessel is needed, which is much more difficult.

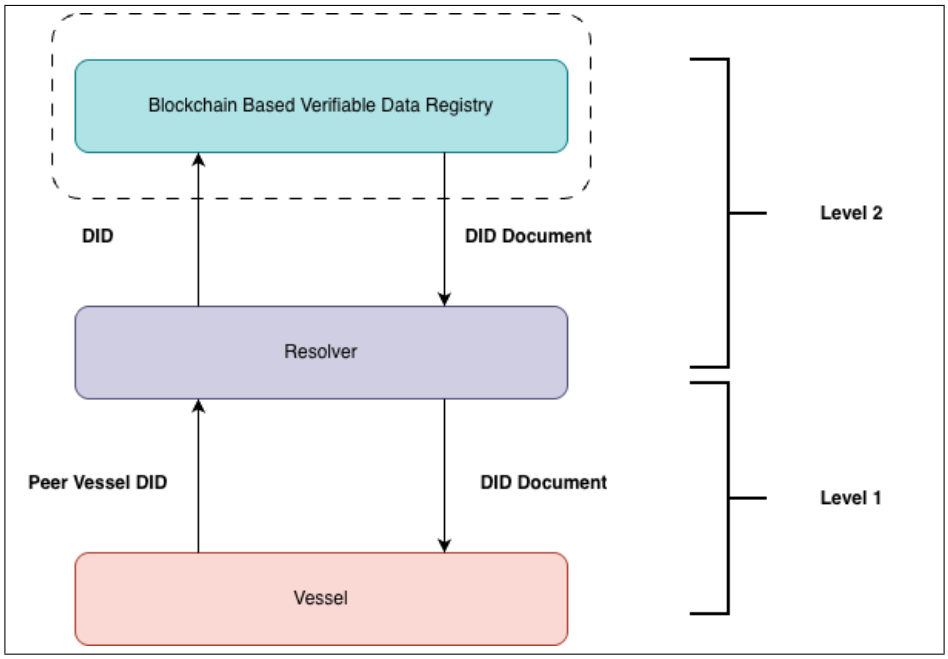


Fig. 3. DID Resolution for a Vessel DID

Vessel DID Registration/Revocation A new vessel joining the fleet generates its public key and private key pair on its own. The vessel also forms its DID from the generated public key. This DID uniquely identifies the vessel in the ocean. Whereas the private key is used for signing the transmitted data. A competent authority such as the shipping company is responsible for the registration of the DID of the concerned vessel. The authority forms a DID document, as shown in Fig. 4 containing characteristics of the vessel. The DID document is bound to the DID of the vessel. The registration is done on some trusted shared platform, such as the blockchain. The sequence diagram for vessel registration is shown in Fig. 9. Similarly, the concerned shipping company is entrusted with vessel DID revocation.

Vessel DID Resolution A vessel DID needs to be resolved to obtain information specific to the vessel. A vessel receiving a vessel DID from its peers resolves the DID.

We introduce a two-layer approach for DID resolution, as shown in Fig. 3. At Layer 1, vessels maintain a local datastore of known vessel DID's and their corresponding DID document. When a vessel DID arrives, it is searched in the local datastore with the help of a resolver. If no entry is found in the local datastore, a query is made to the public Verifiable Data Registry (VDR). This comes in handy when the vessel is operating in the middle of the ocean where proper

communication infrastructure is unavailable. Vessels can sync periodically or opportunistically to the Layer 2 VDR to update the state of the local datastore.



```
{
"@context": ["https://www.w3.org/ns/did/v1"],
"id": "did:vessel:0xb96ab6f1A9a20276512F731A455Ed21784eEcb5c",
"controller":"did:vessel:0x14149d43c82f08a3bb5a07586A00a55126A41107",
"verificationMethod":[{
"id": "did:vessel:0xb96ab6f1A9a20276512F731A455Ed21784eEcb5c#key-ais-1",
    "type": "JsonWebKey2020",
    "publicKeyJwk" : {"kty'": 'EC', "crv" : "P-256"}
}],
"service": [
    {"type": "RevocationStatus", "endpoint" : "https://registry.vessel.eu/..."}
],
"authentication" : [
    "did:vessel:0xb96ab6f1A9a20276512F731A455Ed21784eEcb5c#key-ais-1"
]
}
```


Fig. 4. Vessel DID Document

5.5 Design Goals

The three main design goals of the paper are as follows:

Functional Goal The approach aims to provide a privacy-preserving authentication mechanism for standard AIS messages. Unlike MMSIs, which are reused by different vessels, our DID-based vessel identity is unique throughout the system. Furthermore, the work extends to a secure data delivery mechanism for offshore vessels. The data delivery mechanism relies on secure AIS messages.

Security Goal The approach should provide secure authentication and tamper-proof data delivery.

Privacy Goal

1. Privacy guarantees for peer-to-peer: A vessel should not learn any details about its peer besides its DID and operation status.
2. Privacy guarantees for file download: The centralised artefact server, in our case the Galileo GSC, should not be able to identify an individual vessel.

The approach should not leak any vessel-associated private information.

5.6 Attack Model

We assume there are active as well as passive attackers in the system. The attacker listens to the communications happening in and around it, including the ones in which the adversary is participating. The attacker is eager to learn from the communication and also modify the messages being communicated for personal gains.

6 Proposed Scheme

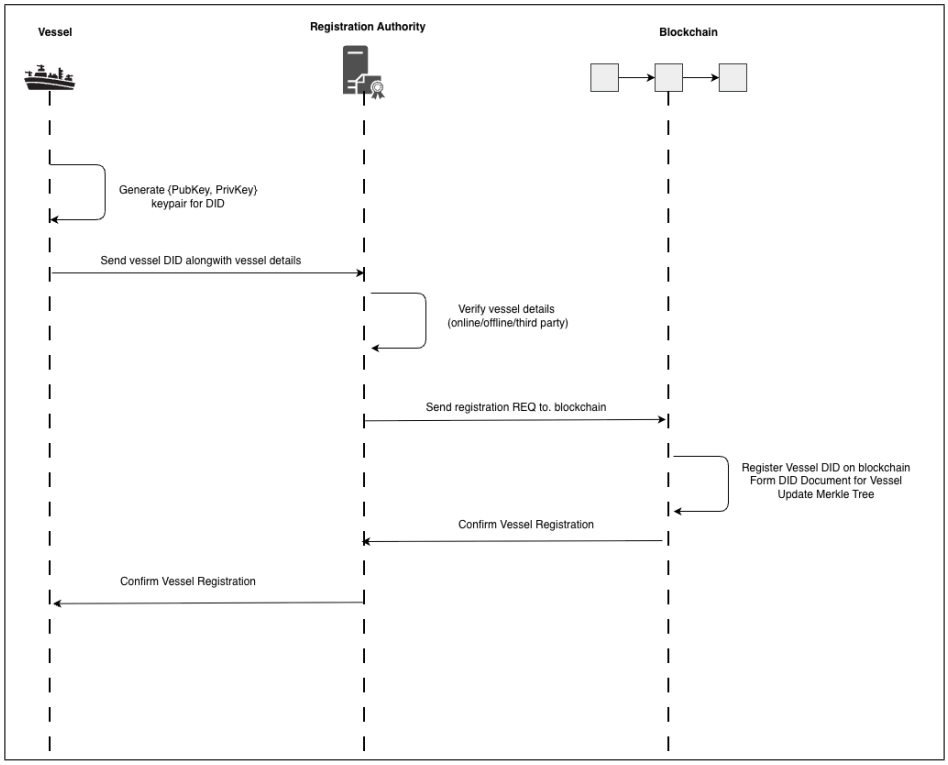


Fig. 5. Sequence diagram for vessel registration

This section discusses the setup, signature, verification, and data delivery mechanism of our approach in detail.

6.1 Setup

The setup procedure for our system is shown in Fig. 5. A vessel generates a public-private key pair. It then derives a DID from its public key and approaches

the Registration Authority (RA) for registration before going on operating in the sea. RA is considered to be a trusted authority in the system. The vessel submits its public key and DID along with other details to the RA. The RA verifies the vessel details, such as the owner details, operating conditions, insurance, etc. Once the due diligence is made and if the vessel satisfies the ready-to-operate conditions, the RA makes an entry of the vessel's DID and public key in the VDR, adds the vessel DID in the Merkle tree as a leaf node, and subsequently updates the Merkle tree. It then signs the necessary details of the vessel and delivers it to the vessel as a credential.

6.2 Sign and Send AIS Data

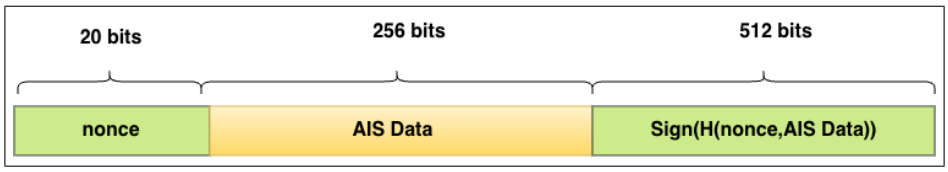


Fig. 6. Message structure of Authenticated AIS

The vessel generates necessary information to be transferred as AIS messages. For this it uses several different components onboard, such as the GNSS receiver to capture signals from space. Once the information pertaining to the AIS message is gathered, the vessel adds a nonce value, then hashes the AIS message along with the nonce and digitally signs the message with its private key. Then the vessel broadcasts the message. The message structure considered in our work is as shown in Fig. 6.

The vessel also shares the path from the leaf node where data is the vessel DID and the adjacent leaf node of the vessel DID leaf node, which is used together with the vessel DID inorder to calculate the hash at the upper layer (i.e., Layer 1). to the Merkle root. A pictorial representation of this can be seen in Fig. 7

6.3 Verify AIS Data

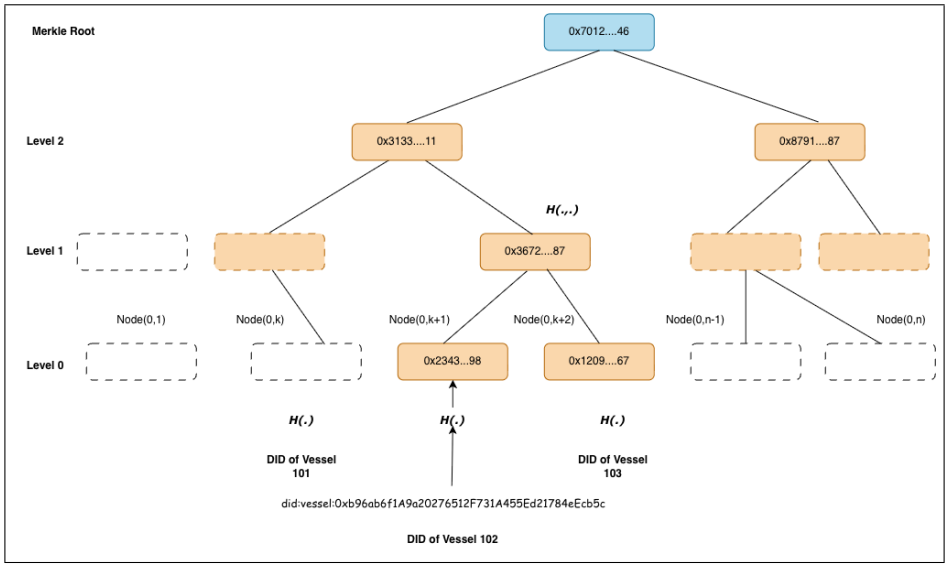


Fig. 7. Diagram depicting the Vessel DID inclusion in the Merkle Tree

The peer vessel, upon receiving the AIS message from the sender vessel, verifies the signature. The signature should match against the sender's vessel DID. Once the signature is matched by the verifier then hashes the AIS message and compares it against the hash sent by the sender vessel.

The vessel also checks the validity of the DID in order to determine weather the vessel DID is revoked or not. The check is made by tracing the Merkle path from the vessel DID leaf node to the Merkle root. The algorithm to check vessel DID inclusion is provided in Algorithm 1.

Algorithm 1 Merkle Tree Verification

```
procedure VerifyMerkleProof(leaf, proof, index, root)
    Input:
        leaf ∈ Bytes                                  ▷ The data block being verified
        proof ∈ List[Bytes]          ▷ Ordered array of sibling hashes from leaf to top
        index ∈ Integer                  ▷ 0-based position of the leaf in the tree
        root ∈ Bytes                                    ▷ The trusted Merkle root
    Output: Boolean                          ▷ True if valid, False otherwise

            ▷ Step 1: Compute initial leaf hash using leaf domain separation (0x00)
    currentHash ← Hash(0x00 ‖ leaf)
                        ▷ Step 2: Iteratively climb the tree using the proof elements
    for each sibling in proof do
        if index (mod 2) == 0 then
                                ▷ Index is even: currentHash is left child, sibling is right
            currentHash ← Hash(0x01 ‖ currentHash ‖ sibling)
        else
                                ▷ Index is odd: currentHash is right child, sibling is left
            currentHash ← Hash(0x01 ‖ sibling ‖ currentHash)
                                    ▷ Move to the parent level using integer division
        index ← ⌊index/2⌋

                ▷ Step 3: Check if the final computed hash matches the trusted root
    return (currentHash == root)
```

6.4 Peer-to-Peer Data Delivery

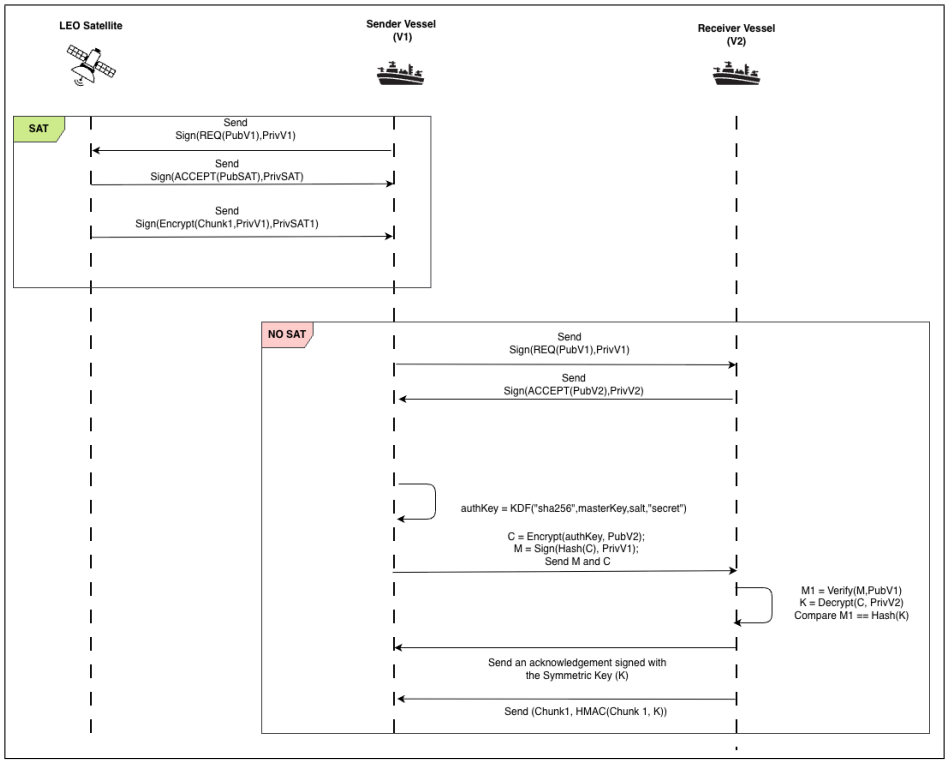


Fig. 8. Sequence diagram for message delivery; satellite-assisted and peer-to-peer

1. Satellite Assisted: When connectivity to the LEO satellite exists, the vessel in need of downloading the RECS file makes a request through the satellite. The satellite authenticates the incoming requests and delivers the artifacts. The sequence of steps performed for satellite-assisted data delivery is shown in Fig. 8.
2. Vessel to Vessel: In case of unavailability or poor connectivity of satellite communications, either because of subscription issues or weather conditions, the vessel can request for the latest RECS artifacts from its nearby peer vessels, which possess the recent RECS artifacts. Like the satellite-assisted data transfer, both vessels first authenticate themselves before initiating data transfer. Further details of vessel-to-vessel data transfer is provided in Fig. 8.

Once the received AIS message is verified, a vessel can be assured of the legitimacy of its peer. A vessel in need of downloading a RECS file can now request

its peers. The requesting vessel periodically broadcasts request messages. Peer vessels with a valid RECS file can reply to the message. Because of the overheads of public key encryption, we propose to use shared key encryption for the purpose of encrypting the RECS file chunks. The algorithm for data transmission and data verification is provided in Algorithm 2 and 3 respectively.

Algorithm 2 Sign and transmit the RECS file.

```
1: procedure SignandTransmit(*file, K)                    ▷
2:     chunks ← SPLIT(file)
3:     while r ≠ Len(chunks) do                           ▷
4:         data ← chunks[r]
5:         tag ← HMAC(data, K)
6:         Send(data, tag)
7:     return true                                        ▷
```

Algorithm 3 Receive and Verify RECS file

```
1: procedure ReceiveAndVerify(*file, K, tag)
2:     fileBuffer ← List()
3:     while socket.Recv() ≠ EOF do                       ▷
4:         data ← socket.Recv()
5:         b ← HMAC(data, K)
6:         if tag == b then
7:             fileBuffer.append(data)
8:     return fileBuffer                                  ▷
```

7 Security Analysis

1. Non-repudiation: Non-repudiation is a mechanism to prevent a party taking part in a communication exchange from denying sending a message later on.
   (a) Non-repudiation in AIS Message Delivery: The non-repudiation is achieved with the notion of use of DID. As each AIS message is digitally signed with the private key associated with the DID whose sole possession lies with the vessel.
   (b) Non-repudiation in file transfer: When a vessel transfers a file to its peer it calculates the HMAC on the data to be transferred and sends it alongside the plaintext data. The secret key used to derive the HMAC is securely shared with the existing PKI.
2. Anonymous Authentication: DID is a pseudonym which exposes no Vessel Identifiable Information (VII). The mapping of a DID to its VII is only known to the registration authority which is a trusted authority in the system.

3. Efficient data transmission: Use of symmetric key cryptography and HMAC for file transfer provides efficient data transmissions.
4. Replay-resistant AIS messages: The *nonce* field introduced in this work provides immunity against replay attacks in the system.

8 Experiment and Results

In this section we discuss the implementation and performance of our proposed approach. We simulated the experiment on a personal computer, a MacBook Pro M2 with 8 GB of RAM and 256 GB of SDD. Both the sender and the receiver side of the AIS message were simulated.

The results from the experiment are as follows:

1. Cryptographic Latency: We measured the signature generation and verification time. Fig. 9 shows the cryptography latency plot.

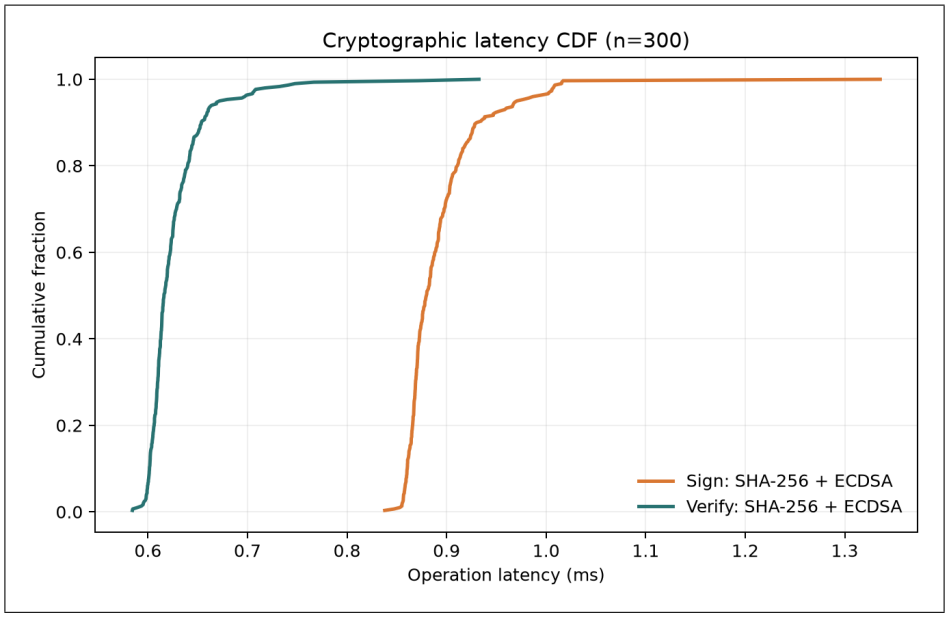


Fig. 9. Signature Generation and Verification

2. Message Size Comparison:

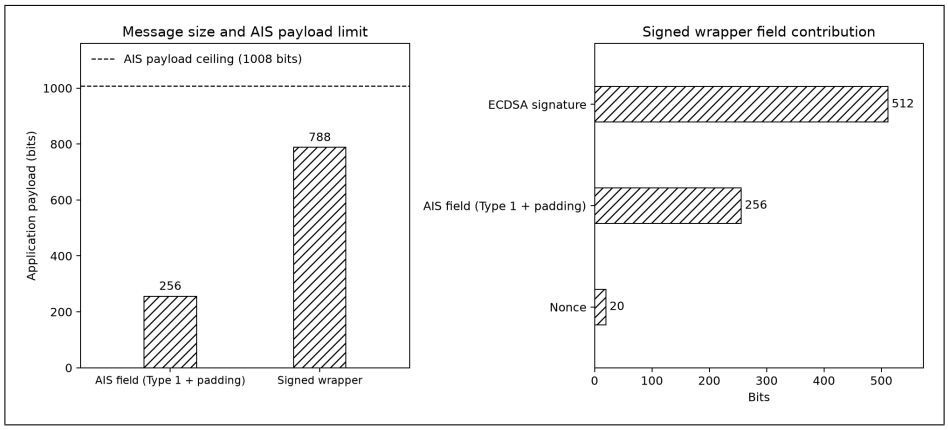


Fig. 10. Payload Overhead

The payload overhead associated with our work is compared as shown in Fig. 10. Although our approach has a higher payload in comparison to the plain AIS, it's still below the AIS payload limit.

3. Channel Load Comparison:

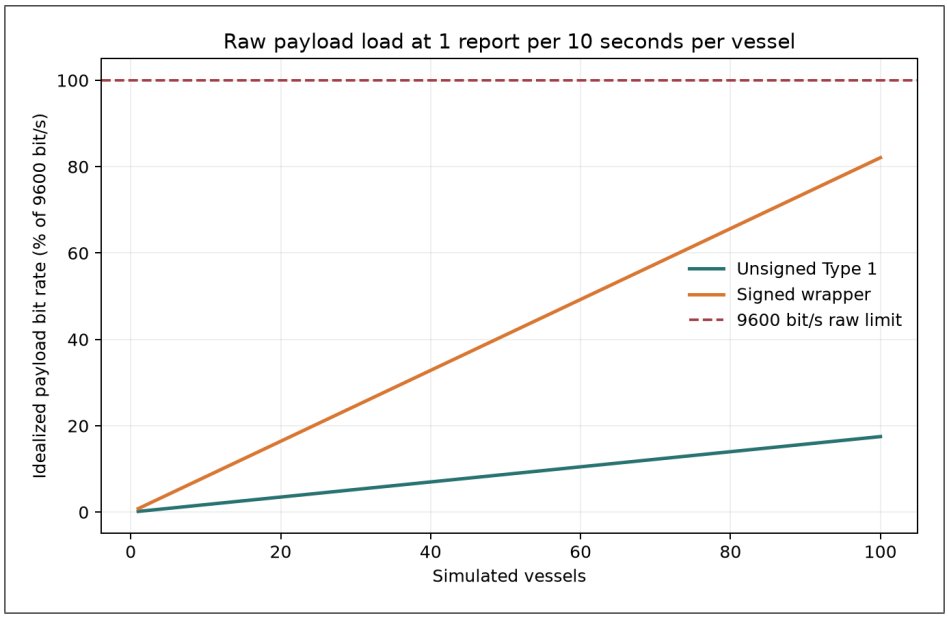


Fig. 11. Channel Load Comparison

Method	Response Time
Sepolia Testnet	5.2 sec.
BadgerDB	0.009 ms.
Cache	0.006 ms.

Table 2. Response Time

We implemented the VDR on the Sepolia testnet, a public Ethereum test network. The VDR was implemented as a smart contract. The Certificate Authority deploys the VDR smart contract on the blockchain. The smart contract implements the data structure used to form the DIDDocument. It also exposes a Getter function to retrieve the records needed to form the DIDDocument. Information For the local data registry we used badgerDB. We measured the response delay in three scenarios. Table 2 provides the response time delay for resolving a DID to a DID document against different platforms. Moreover, the DIDDocument size was found to be 571 bytes.

8.1 Algorithmic Analysis of Merkle Tree Verification

Merkle tree verification consists of two main operations:

1. Leaf Search: Locate the index of the leaf node in the tree.
2. Inclusion Proof Computation: Compute the Merkle tree root from the leaf node.

However, in our case the leaf search part is relieved as the sender sends the subtree from the concerned leaf tree to the Merkle tree root, as not all vessels can efficiently store the entire tree, as it needs frequent updates. The only task in hand is to compute the inclusion proof. The worst-case complexity of inclusion proof checking is $O(\log n)$, where n is the number of leafs in the tree.

9 Discussion

In this section, we discuss and analyse the achivements of the work.

1. Location validity check: The location validity is assured with the use of Signal Authentication Service from Galileo besides the OSNMA service. OSNMA guarantees the signal is generated from a valid Galileo satellite where SAS guarantees that the signal is really directly transmitted from the satellite and not replayed from any other source.
2. Order check for AIS messages: The work introduced a *nonce* field in order to distinguish between two AIS messages. A receiver always waits for an AIS message with a nonce greater than the present nonce available for a AIS transmitter. Any AIS message with a lower nonce available is considered as stale and is discarded. This handles the replay attack.

3. Authenticated AIS: AIS transponders are provided with a DID. Each AIS message transmitted by a transponder is digitally signed by the transponder's private key.
4. Integrity check for AIS messages: The integrity of an AIS message is preserved as the transponder digitally signs the AIS message hash. At the receiving end, the AIS receiver hashes the AIS message and compares it with the digitally signed hash.
5. Enhanced LEO satellite service management: As both the satellites and vessels are registered on the blockchain with uniquely identifiable DIDs, it allows a satellite provider to efficiently manage the subscriptions.

10 Conclusion

Secure data delivery in the open ocean is a challenge. The messages relayed are available for capture, modification, and replay by any adversary. The same applies to AIS, an information-sharing protocol in maritime. In this paper, a secure and privacy-preserving approach for data delivery in offshore vessels is proposed. The work uses DID for vessel identification and authentication. DID provides privacy to the vessel, as it does not disclose identification details of the vessel beyond the pseudonym. Moreover, the vessels can now securely share relevant data as per the need. The vessel DID resolve to a DID document, which tells the peer vessels the way to interact with it.

The work further shows how the enhanced AIS message can be used to deliver artefacts to the vessels. It considers the use case of RECS file transfer. The vessel uses a symmetric key for data encryption during transmission. The symmetric key is shared using the DID keys. The approach is implemented, and performance is measured. The channel load is within the 9600 baud rate limit for AIS message authenticated transmission.

References

1. Arifin, M.D., Muslim, M., Octaviani, F., Faturachman, D.: Review of automatic identification system: Application, challenge and limitations. Int. J. Mar. Eng. Innov. Res **9**(3), 464–475 (2024)
2. Authentication, B.C.: Auth-ais: Secure, flexible, and backward-compatible authentication of vessels ais broadcasts (2021)
3. Balduzzi, M., Pasta, A., Wilhoit, K.: A security evaluation of ais automated identification system. In: Proceedings of the 30th annual computer security applications conference. pp. 436–445 (2014)
4. Duan, J., Jiang, S., Guo, L., Zhang, J.: An ais-based routing protocol for extending shore-based networks to near-shore water areas. Ocean Engineering **341**, 122562 (2025)
5. EUSPA: Galelio acas (2026), "https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/tender-details/docs/etender/14288/.../GalileoACASv1.pdf"

6. EUSPA: Hyperlinks in latex (2026), https://www.newindianexpress.com/india/2025/Dec/11/over-1900-incidents-of-gps-spoofing-of-aircraft-reported-in-two-years-centre-tells-lok-sabha
7. EUSPA: Osnma (2026), https://www.gsc-europa.eu/galileo/services/galileo-open-service-navigation-message-authentication-osnma
8. EUSPA: Rfc (2026), https://datatracker.ietf.org/doc/html/rfc4082
9. EUSPA: Spoofing in middle east (2026), https://windward.ai/blog/gps-jamming-disrupts-1100-ships-in-the-middle-east-gulf/
10. Fernandez-Hernandez, I., Winkel, J., O'Driscoll, C., Willems, T., Cancela, S., Ramirez, M.A., Terris-Gallego, R., Lopez-Salcedo, J.A., Seco-Granados, G., Fuchs, F., et al.: Prototyping galileo signal authentication service: Current status and plans. Engineering Proceedings **126**(1), 40 (2026)
11. GNSS, I.: Galelio outage (2026), https://insidegnss.com/galileo-outage-update-service-failure-poses-serious-questions/
12. Goudosis, A., Katsikas, S.: Secure automatic identification system (secais): Proof-of-concept implementation. Journal of Marine Science and Engineering **10**(6), 805 (2022)
13. Goudosis, A., Katsikas, S.K.: Secure ais with identity-based authentication and encryption. TransNav: International Journal on Marine Navigation and Safety of Sea Transportation **14**(2), 287–298 (2020)
14. Ji, M., Cui, X., Li, J., Zhang, H., Li, S., Liu, J.: An opportunistic network routing protocol for maritime voluntary observation ships. Procedia Computer Science **202**, 390–393 (2022)
15. Kessler, G.C., Zorri, D.M.: Ais spoofing: A tutorial for researchers. In: 2024 IEEE 49th Conference on Local Computer Networks (LCN). pp. 1–7. IEEE (2024)
16. Lenhart, M.: Distributed relay/replay attacks on gnss signals (2022)
17. Ray, C., Gallen, R., Iphar, C., Napoli, A., Bouju, A.: Deais project: Detection of ais spoofing and resulting risks. In: OCEANS 2015-Genova. pp. 1–6. IEEE (2015)
18. Su, P., Sun, N., Zhu, L., Li, Y., Bi, R., Li, M., Zhang, Z.: A privacy-preserving and vessel authentication scheme using automatic identification system. In: Proceedings of the fifth ACM international workshop on security in cloud computing. pp. 83–90 (2017)
19. Suárez, P.G.: Python implementation of an anti-replay method for galileo osnma using sample-level partial-correlation metrics. arxiv (2026)
20. Wang, H., Zhang, Y., Shen, Y., Zhu, J., Chen, Y., Jiang, X.: Novel replay attacks against galileo open service navigation message authentication. In: Proceedings of the 36th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2023). pp. 3897–3907 (2023)
21. Yang, Q., Laurenson, D.I., Barria, J.A.: On the use of leo satellite constellation for active network management in power distribution networks. IEEE Transactions on Smart Grid **3**(3), 1371–1381 (2012)